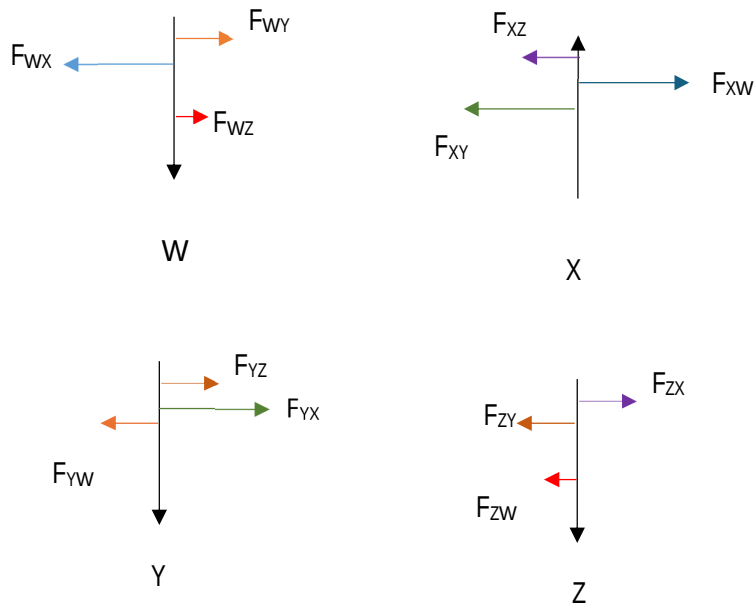


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Answer: C

Draw the magnetic forces acting on each wire due to the rest of the wires.

Wire Y has a large magnetic force acting to the right due to wire X and its larger current. Wire Y is also attracted to the right by wire Z.



OR

Force on wire X due to wire Y = F_{XY}

$$F_{XY} = B_Y I_X L$$

$$B_Y = \frac{\mu_0 I_Y}{2\pi a}$$

$$F_{XY} = \frac{\mu_0 I_Y}{2\pi a} I_X L$$

$$F_{XY} = \frac{\mu_0 I_Y I_X L}{2\pi a}$$

$$= k \frac{I_Y I_X}{a} \text{ where } k = \frac{\mu_0 L}{2\pi}$$

Taking to the right as positive:

$$F_W = -F_{WX} + F_{WY} + F_{WZ}$$

$$= -k \frac{2I^2}{a} + k \frac{I^2}{2a} + k \frac{I^2}{3a}$$

$$= -k \frac{7I^2}{6a}$$

$$\begin{aligned}
 F_x &= -F_{xy} - F_{xz} + F_{xw} \\
 &= -k \frac{2l^2}{a} - k \frac{2l^2}{2a} + k \frac{2l^2}{a} \\
 &= -k \frac{l^2}{a}
 \end{aligned}$$

$$\begin{aligned}
 F_y &= -F_{yw} + F_{yx} + F_{yz} \\
 &= -k \frac{l^2}{2a} + k \frac{2l^2}{a} + k \frac{l^2}{a} \\
 &= k \frac{5l^2}{2a}
 \end{aligned}$$

$$\begin{aligned}
 F_z &= -F_{zw} - F_{zy} + F_{zx} \\
 &= -k \frac{l^2}{3a} - k \frac{l^2}{a} + k \frac{2l^2}{2a} \\
 &= -k \frac{l^2}{3a}
 \end{aligned}$$